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Miocene Paleomagnetism and Tectonic Setting of the Baja California Peninsula, Mexico

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A paleomagnetic pole calculated from 64 virtual geomagnetic poles for Miocene volcanic rocks of the Baja California Peninsula (86.1°N, 142.6°E, α_{95} = 3.9°) is similar to a reference pole for the North American craton (87.4°N, 129.7°E, α_{95} = 3.0°), consistent with the Baja Peninsula having been adjacent to the Mexican mainland in its pre-Gulf of California position from at least early Miocene time to the Pliocene opening of the Gulf. This result is in agreement with marine geophysical and geologic evidence from the Gulf of California and Baja Peninsula and does not support a long-period nondipole geomagnetic field in the region during Miocene time, or 10° of northward transport of the peninsula since then, inferred from smaller data sets (included in this work) by previous workers. This study also indicates that a pre-Gulf of California northward translation (9° of latitude) and clockwise rotation (24°) of the Baja Peninsula with respect to North America, determined from paleomagnetic studies of Cretaceous rocks within peninsular California occurred prior to Miocene time. Paleomagnetic results from Late Cretaceous to early Tertiary basement rocks in western Sinaloa, Mexico are similar to those results from Cretaceous rocks of the Baja Peninsula, supporting the early Tertiary northward translation of the peninsula between 45 and 34 Ma and suggesting that the tectonic boundary separating peninsular California from North America at that time was east of the present-day Gulf of California.

INTRODUCTION

Marine geophysical investigations near the mouth of the Gulf of California [*Larson*, 1972; *Mammerickx*, 1980] have shown that the Baja California Peninsula (Figure 1) has been rifted northwestward (≈2° of latitude) and rotated clockwise (≈8°) along the Pacific-North American plate boundary from a position contiguous with the Mexican mainland since late Miocene time. Correlations of Miocene volcanic and sedimentary units across the Gulf of California [*Gastil et al.*, 1979; *Hausback*, 1984*b*] also indicate that the Baja Peninsula was adjacent to western Mexico in early Miocene time. Geologic correlations made between Mesozoic rocks of the Baja Peninsula and Sonora, Mexico [e.g., *Silver et al.*, 1979] suggest that no translations of the peninsula beyond the opening of the Gulf of California have occurred since Late Cretaceous time. However, the amount of northward translation (11° of latitude) and clockwise rotation (32°) of the Baja Peninsula estimated from paleomagnetic studies of Cretaceous igneous and sedimentary rocks within the peninsula [*Hagstrum et al.*, 1985; *Filmer and Kirschvink*, 1986] exceeds that associated with the opening the Gulf of California. Most of this motion since the Late Cretaceous (9° of northward translation, 24° of clockwise rotation) had to have occurred prior to early Miocene time in order to place the Baja Peninsula in its pre-Gulf position along the western Mexican mainland. This early Tertiary displacement is consistent with paleomagnetic data from many other Cordilleran terranes [*Coney et al.*, 1980] that indicate large northward translations and clockwise rotations relative to cratonic North America during much of Tertiary time [*Beck*, 1976, 1980].

In conflict with the geologic and geophysical evidence, *Pischke et al.* [1979] and *Marshall et al.* [1979] concluded from paleomagnetic studies of Neogene volcanic rocks from the Baja Peninsula that the peninsula was 10° south of its present position in late Miocene time (see also *Pischke* [1979]). *Luyendyk et al.* [1985] also noted shallow inclinations in some Miocene volcanic rocks from southern California and western Arizona and proposed that the anomalous inclinations could be explained by a long-period nondipole geomagnetic field in southwestern North America during Miocene time. Recently, *Pischke et al.* [1986] suggested that an anomalous field, instead of tectonic translation, also could be responsible for the low inclinations observed in Cretaceous rocks from the Baja Peninsula [e.g., *Hagstrum et al.*, 1985]. In reply to this suggestion, *Hagstrum et al.* [1987] pointed out that an anomalously shallow-inclination field in southern and peninsular California is indicated neither by the available paleomagnetic data for other Cretaceous rocks of western North America nor by those, mostly unpublished, for Miocene rocks from the Baja Peninsula.

The purpose of this paper is to present the available paleomagnetic data for Miocene rocks from the Baja California Peninsula generated by the authors of this study and

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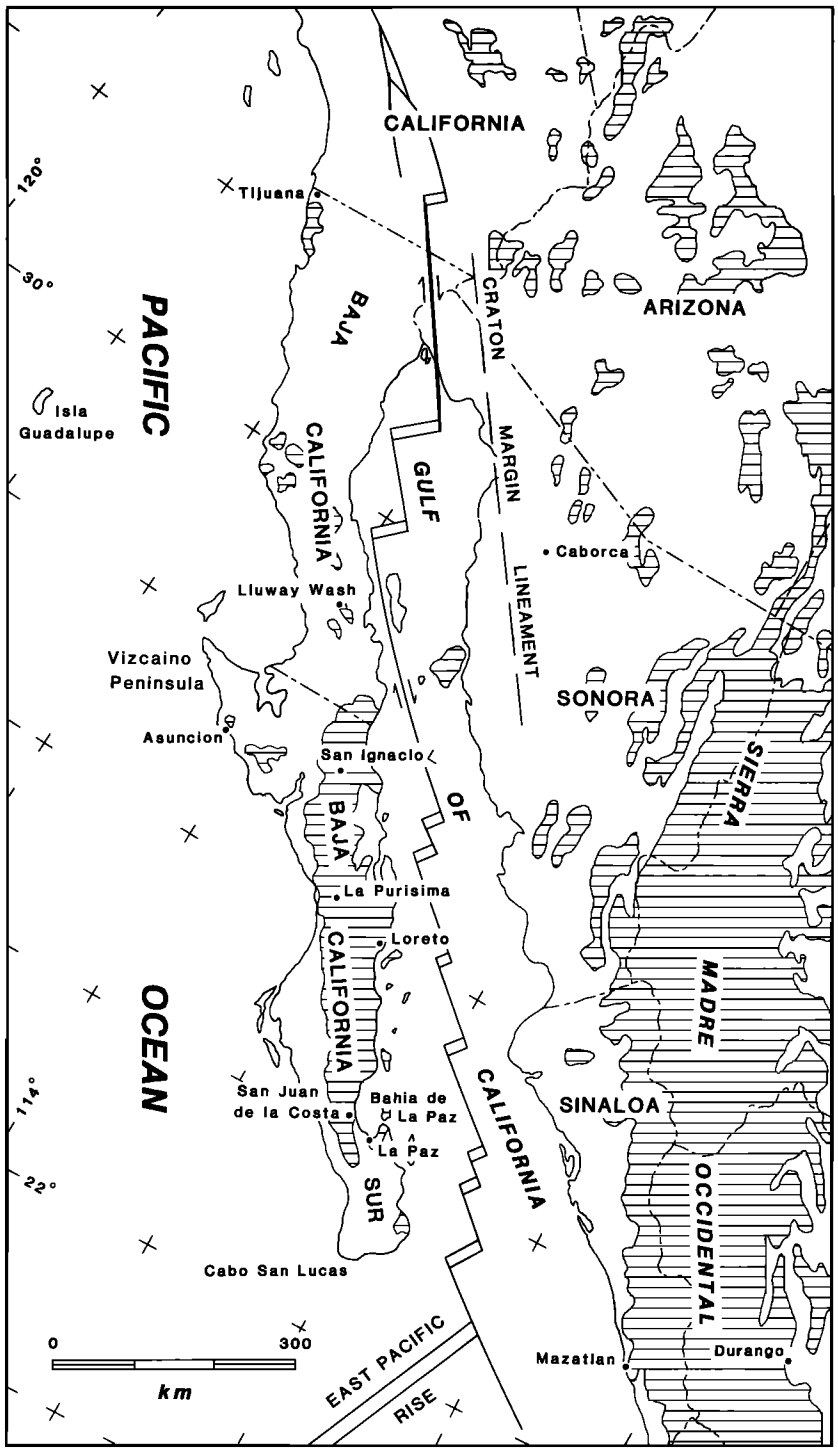


Fig. 1. Index map of the Baja California Peninsula and western Mexican mainland showing Tertiary volcanic rocks (shaded areas) and regional tectonic features [after *Goddard et al.*, 1965; *Gastil and Krummenacher*, 1977].

including those of *Pischke* [1979] and *McNaboe* [1986]. This extensive database does not support a long-period nondipole geomagnetic field during Miocene time and is consistent with the Baja Peninsula being adjacent to the Mexican mainland from at least early Miocene time to the Pliocene opening of the Gulf of California.

Regional Geology and Tectonics

The paleomagnetic results presented in this study are from volcanic rocks which span Miocene time (Table 1) and reflect the changing tectonic regimes affecting the Baja California Peninsula. Prior to Miocene time, the Farallon plate was being subducted beneath southwestern North America [*Atwater*, 1970; *Mammerickx and Klitgord*, 1982], and an associated calc-alkaline volcanic arc was active in western Mexico [*Clark et al.*, 1982]. A major episode of rhyolitic volcanism occurred in the Sierra Madre Occidental between 34 and 27 Ma and its products were emplaced unconformably over an older sequence (102 to 45 Ma) of generally intermediate-composition volcanic and related

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TABLE 1. Site Mean Paleomagnetic Directions and Poles for the Baja California Peninsula, Mexico

Site	Slat	Slong	Age	I	D	N	R	k (k_1/k_2)	α_{95} (α_1/α_2)	Plat	Plong
Sites Near La Purisima, Baja California Sur											
LP1	26.04	247.80	8.52*	45.2	334.1	7	6.9235	78	6.9	66.7	165.2
LP2	26.33	248.10	7.65*	54.9	359.3	7		−2031/−1	1.1/1.4	80.9	244.5
LP3	26.33	248.11	7.6	50.5	352.2	6		−281/−1	3.1/3.8	81.6	195.5
LP4	26.35	248.15	8–14	49.1	352.4	6		−134/−1	4.7/5.3	82.4	188.1
LP5	26.35	248.14	8–14	37.4	358.4	3		−30/−1	13.4/18.6	84.4	83.3
LP6	26.36	248.29	8–14	52.2	352.2	7		−27/−4	8.0/21.6	80.6	203.5
LP7†	26.37	248.28	17.8*	71.1	185.0	6	5.9215	64	8.5	−7.9	245.4
LP8†	26.37	248.28	17.8*	66.6	185.6	6	5.9146	59	8.8	−14.4	244.5
LP9	26.26	248.03	8.93*	45.2	340.1	7	6.9827	346	3.3	72.2	164.1
LP10	26.19	248.08	6–9	50.3	348.6	6		−73/−5	5.1/14.3	78.9	186.7
LP11	26.19	248.03	6–9	49.8	340.5	6		−877/−1	1.7/2.2	72.3	177.1
LP12	26.36	248.16	8.51*	54.9	7.4	7	6.9884	516	2.7	78.9	281.3
LP13	26.36	248.16	10	−42.9	187.3	8	7.9852	474	2.6	83.3	348.8
LP14	26.36	248.16	13.7*	−49.4	161.0	5	4.9987	2960	1.4	72.9	175.8
LP15	26.38	248.27	16–18	−64.2	180.5	5	4.9477	76	8.8	70.5	149.3
LP16	26.38	248.27	16–18	−73.3	160.8	7	6.9921	756	2.2	54.8	131.2
LP17	26.35	248.22	10.8*	−21.1	163.4	5	4.9939	656	3.0	68.0	116.8
LP18	26.35	248.24	11.0*	−47.7	176.3	4	3.9373	48	13.4	85.9	196.0
LP19†	26.37	248.28	17.8*	67.5	163.4	7	6.9867	452	2.8	−11.8	259.0
LP20	26.37	248.28	17.8*	38.4	11.3	5		−68/−2	6.6/9.7	78.7	0.6
LP21	26.35	248.24	11.0*	−29.1	156.2	7	6.9701	201	4.3	65.4	137.0
LP22	26.20	248.02	5–7	−57.7	156.0	7	6.9911	674	2.3	66.5	195.0
LP23	26.24	247.93	5–7	−50.2	212.7	7	6.9850	401	3.0	61.0	320.8
LP24	26.17	247.95	14*	43.6	345.8	6	5.9844	320	3.8	77.2	157.9
LP25	26.29	247.53	7.71*	33.2	351.6	7	6.9914	699	2.3	78.8	112.7
Sites Near San Ignacio, Baja California Sur											
SI1	27.35	247.33	11–14	−33.7	171.3	5	4.9811	212	5.3	78.0	111.1
SI2	27.38	247.29	10	55.1	358.4	8	7.9901	708	2.1	81.6	238.3
SI3	27.41	247.34	11–14	39.4	340.6	7	6.8911	55	8.2	71.7	145.5
SI4†	27.46	247.31	11	27.2	84.0	7	6.9849	397	3.0	11.8	327.1
SI5†	27.46	247.31	11	16.3	83.7	10	9.7785	41	7.8	9.4	332.8
SI6	27.44	247.33	10	52.2	10.5	7	6.9826	345	3.3	79.5	304.2
SI7	27.44	247.33	11	21.2	350.8	10	9.9267	123	4.4	71.4	96.8
SI8†	27.46	247.34	11	12.2	79.8	8	7.9739	268	3.4	11.9	336.5
SI9	27.41	247.25	2.99*	−44.3	164.7	7	6.9922	765	2.2	76.3	154.9
SI11	27.43	247.26	3–5	−41.1	175.1	7	6.9942	1038	1.9	84.1	117.2
SI12	27.44	247.36	11	−53.8	140.1	7	6.9880	498	2.7	55.3	178.9
SI13	27.33	247.22	3–5	−51.7	179.7	5	4.9717	141	6.5	85.0	244.3
SI14a	27.43	247.40	10	51.8	1.0	7	6.9889	540	2.6	84.9	257.0
SI15a	27.43	247.40	10	54.6	0.1	8	7.9807	363	2.9	82.3	248.0
SI16a	27.43	247.40	10	50.6	3.8	8	7.9722	252	3.5	84.9	286.8
a						3	2.9980	1020	3.9	84.3	261.5
SI17b	27.29	247.08	10	25.8	44.4	8	7.9856	485	2.5	46.5	346.2
Sites Near Bahia de La Paz, Baja California Sur											
BP1	24.15	249.21	17.2*	46.7	355.3	7	6.9799	399	2.6	84.3	202.2
BP2	24.15	249.21	17.6*	33.7	4.2	8	7.9816	488	2.2	83.1	34.0
BP3	24.38	249.31	18	26.7	11.8	5	4.9618	157	5.0	74.9	19.9
BP4c	24.24	249.39	20*	−37.9	180.4	7	6.9872	623	2.1	87.0	62.2
BP5	24.45	249.22	21	−33.3	169.0	5	4.9645	169	4.8	78.0	129.9
BP6	24.28	249.37	21.0*	−38.9	140.6	6	5.9754	285	3.4	53.8	163.9
BP7	24.28	249.37	22	13.5	345.3	7	6.9398	133	4.6	67.6	110.7
BP8	24.31	249.36	22	20.7	12.8	7	6.9907	857	1.8	71.7	25.3
BP9	24.31	249.36	22	45.0	359.3	5	4.8947	57	8.3	87.7	233.9
BP10	24.45	249.26	24	26.6	326.9	5	4.7857	28	11.7	57.2	147.0
BP11	24.37	249.28	24	57.2	10.8	5	4.5000	12	18.3	73.7	281.2
BP12	24.11	249.69	19*	36.3	342.8	8	7.9211	114	4.6	73.6	149.2
BP13	24.19	249.70	19.8*	−33.2	186.5	6	5.9825	399	2.9	81.4	23.5
BP14c	24.15	249.73	20*	−35.7	183.5	10	9.9754	447	2.1	84.5	32.6
c						2	1.9992	1245	7.1	85.9	43.0
BP15	24.14	249.73	23.2*	−32.2	173.4	3	2.9897	388	4.1	80.9	113.8
Results From Pischke [1979]											
AM1d	27.30	245.60	8.1*	−35.3	183.3	5	4.9875	320	4.3	81.6	43.7
AM2d	27.30	245.60	13	−39.5	183.7	6	5.9815	271	4.1	84.0	30.2
d						2	1.9994	1540	6.4	82.9	38.1
IM1b	27.30	247.10	10	27.9	45.1	6	5.9946	927	2.2	46.4	344.2
IM2b	27.30	247.10	10.8*	25.7	46.0	5	4.9980	2041	1.7	45.0	345.4
IM3b	27.30	247.10	11	17.3	41.4	6	5.9557	113	6.3	46.8	354.4
IM4b	27.30	247.10	11	20.5	47.6	6	5.9957	1166	2.0	42.3	348.1
IM5b	27.30	247.10	11	21.9	44.7	6	5.9380	81	7.5	45.2	349.9
b						6	5.9928	693	2.6	45.4	348.0
IM6e	27.30	247.10	5	31.4	359.8	7	6.9582	143	5.1	79.7	68.2
IM7e	27.30	247.10	5.1*	35.0	358.2	5	4.9832	238	5.0	81.8	79.1

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TABLE 1. (continued)

Site	Slat	Slong	Age	I	D	N	R	k (k_1/k_2)	α_{95} (α_1/α_2)	Plat	Plong
IM8e	27.30	247.10	5	30.3	358.1	6	5.9888	447	3.2	78.9	76.6
e						3	2.9990	2065	2.7	80.2	74.4
JM1	25.80	248.50	11.0*	−14.6	177.4	6	5.9210	63	8.5	71.5	76.6
JM2	25.80	248.50	14.3*	−10.3	217.9	7	6.8749	48	8.8	48.3	1.6
JM3	25.80	248.50	18.7*	−35.8	206.7	8	7.9118	79	6.3	64.7	346.6
JM4	25.80	248.50	19	−17.9	184.4	5	4.9763	169	5.9	72.9	53.6
JM5	25.80	248.50	7.6*	57.1	352.8	6	5.9752	202	4.7	76.6	223.1
LM1†	24.30	249.70		6.8	30.0	5	4.8999	40	12.3	54.3	10.8
LM2	24.30	249.70	9.0*	−38.0	181.7	7	6.9715	210	4.2	86.6	41.4
LM3	24.30	249.70	14.9*	47.1	19.1	6	5.8307	30	12.5	72.4	322.4
OM1	29.10	246.00	14.4*	−35.4	179.3	7	6.8748	48	8.8	80.4	250.0
OM2f	29.10	246.00	13	62.0	0.3	9	8.9876	644	2.0	75.9	246.8
OM3f	29.10	246.00	12.5*	59.2	358.2	9	8.4343	14	14.2	79.0	238.7
f						2	1.9990	1038	7.8	77.5	243.2
					Results From McNaboe [1986]						
TR1	32.30	243.00	14.3*	−53.6	192.3	7	6.9613	155	4.9	79.5	220.2
TR2	32.30	243.00	14.3*	−56.3	182.4	7	6.9719	214	4.1	85.1	274.3
TR3g	32.30	243.00	14	−50.6	173.2	6	5.9318	73	7.9	84.2	34.5
TR4g	32.30	243.00	14	−50.0	183.6	8	7.9373	112	5.3	86.6	181.0
g						2	1.9941	168	19.4	88.2	66.9
TR5	32.30	243.00	14	−50.8	186.2	7	6.9898	589	2.5	84.7	199.8
TR6	32.30	243.00	14	−58.1	170.8	7	6.9343	91	6.4	80.1	343.4
TR7†	32.30	243.00	14.3*	11.7	146.8	6	5.8692	38	11.0	40.4	71.3

Slat, Slong, latitude and east longitude of sampling sites respectively (in degrees); age, K-Ar age (in Ma); I,D, inclination and declination, respectively (in degrees), corrected for structural tilt; N, number of samples analyzed; R, vector sum; k, precision parameter [*Fisher*, 1953]; α_{95}, radius and 95% confidence in degrees; Plat, Plong, latitude and east longitude of virtual geomagnetic pole, respectively (in degrees). For directions resolved with demagnetization planes and Bingham statistics [*Onstott*, 1980], R is not calculated, k_1/k_2 are the two Bingham concentration parameters, and α_1/α_2 are the two Bingham 95% confidence limits. Letters a–g are averaged sites (see text).

* Indicates age applies directly to unit sampled for paleomagnetism (see text), otherwise the age is an estimate based on dated over- or underlying units or on petrologic and geomorphic comparisons with dated units.

† Sites discarded from mean pole calculation.

batholithic rocks [*McDowell and Keizer*, 1977; *McDowell and Clabaugh*, 1979]. The axis of volcanic activity probably migrated westward and arrived along what is now the eastern margin of the Baja Peninsula by about 24 Ma [*Hausback*, 1984*b*; *Sawlan and Smith*, 1984]. The volcanic rocks deposited in peninsular California range from andesitic flows and lahars to rhyolitic ash-flow tuffs; rhyolitic rocks are abundant in the Bahia de La Paz region (Figure 1) but are sparsely represented elsewhere.

Transform faulting began along the California continental margin at about 29 Ma when the North American plate overrode the Pacific-Farallon spreading center and first contacted the Pacific plate [*Atwater and Molnar*, 1973]. Subduction continued to the south along the western margin of the Baja Peninsula until the northern segments of the Guadalupe-Pacific rise were abandoned [*Mammerickx and Klitgord*, 1982] and the Tosco-Abreojos and San Benito transform faults were initiated at approximately 12.5 Ma [*Spencer and Normark*, 1976]. Extinction of the volcanic activity in peninsular California closely followed the cessation of subduction [*Hausback*, 1984*b*; *Sawlan and Smith*, 1984]. In Middle to Late Miocene time, east-west-directed extension began along the proto-Gulf of California and was characterized in the Baja Peninsula by high-angle normal faulting [*Dokka and Merriam*, 1982; *Colletta and Angelier*, 1983] and widespread predominantly alkalic basaltic volcanism [*Sawlan and Smith*, 1984]. At about 6 Ma the locus of Pacific-North American plate motion shifted from the transform fault(s) off the western coast of the Baja Peninsula to the axis of the Gulf of California [*Mammerickx*, 1980].

FIELD AND LABORATORY METHODS

Samples for paleomagnetic analysis were collected in Baja California Sur (Figure 1) as 2.5-cm drill cores using a portable gasoline-powered drill. The samples were oriented with an estimated accuracy of 2° using a magnetic and/or solar compass, and solely magnetic orientations of the more strongly magnetic basaltic rocks were checked by taking back azimuths. Cylindrical specimens 2.2 cm in length were cut from the core samples and were subjected to either alternating-field (af) or thermal cleaning. Detailed progressive af demagnetizations were usually done on two or three pilot samples from a given site and served to determine the demagnetization steps for the site's remaining samples. Progressive af demagnetization in peak fields between 1 and 100 mT was the primary technique used in this study. Stepwise thermal demagnetizations were performed in air to temperatures usually in excess of 600°C on one or two specimens from sites near Bahia de La Paz and on samples from other sites when af treatments proved ineffective. Single-step af demagnetizations were done on samples from several sites where pilot samples exhibited univectorial magnetizations.

Demagnetization techniques were generally successful in isolating a stable component of magnetization for the rock samples. These components form linear segments on orthogonal vector plots (Figure 2) which decay toward the origin with increased demagnetization. Results for samples having two components with overlapping coercivity spectra described great circle arcs on equal-area plots with increased

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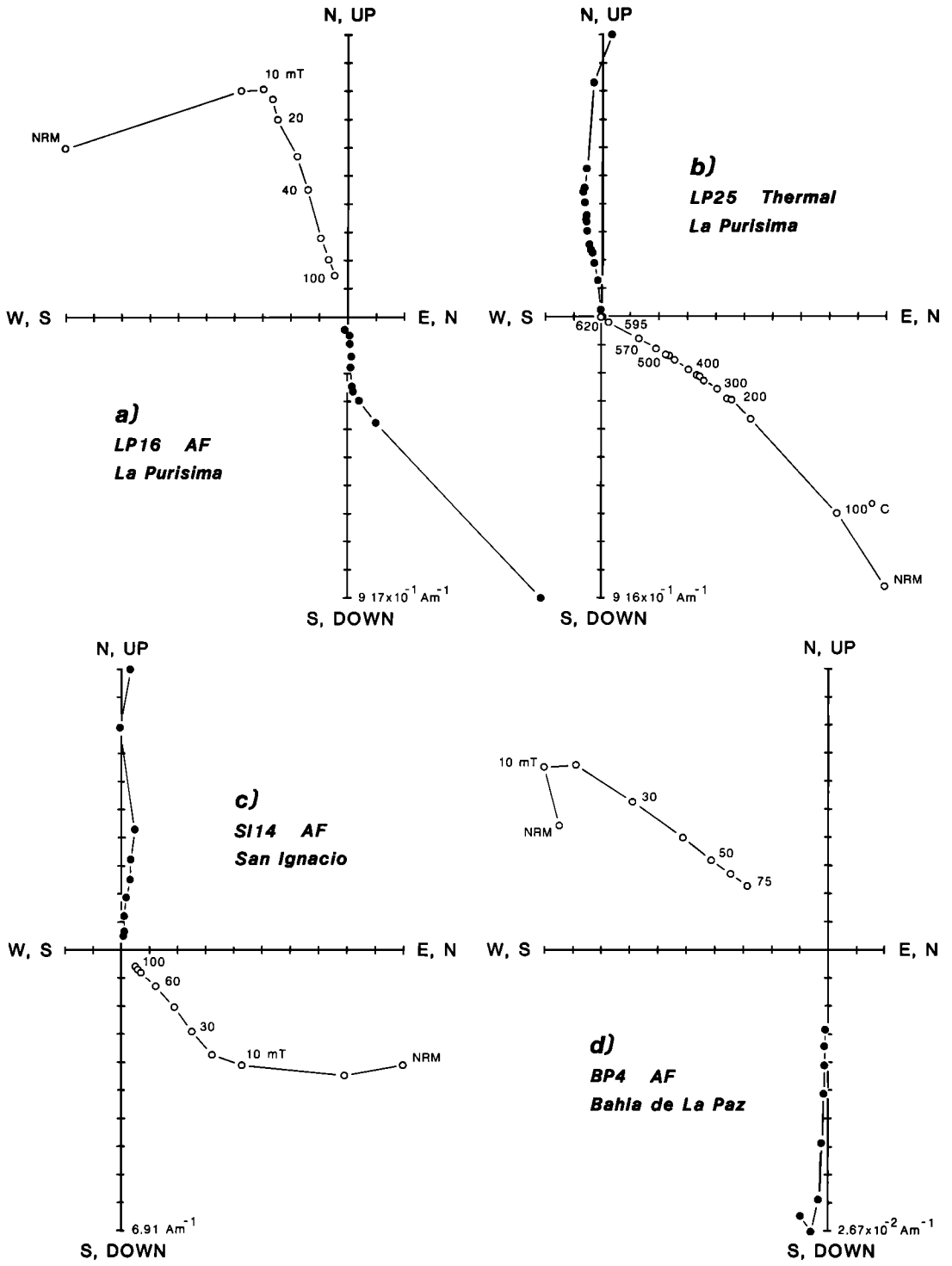


Fig. 2. Orthogonal projections of magnetization vector end points during demagnetization experiments. (*a*) The af results for a sample from site LP16 in a basaltic andesite flow near La Purisima showing the isolation of a stable characteristic magnetization. (*b*) Thermal results for a sample from site LP25 in a basalt flow near La Purisima. (*c*) The af results for a sample from an Esperanza basalt flow (site SI14) near San Ignacio. (*d*) The af results for a sample from the rhyolitic La Paz tuff (BP4) near Bahia de La Paz. Solid circles indicate projections of vector end points onto the horizontal plane and unfilled onto the vertical plane.

demagnetization and required the use of planes analysis [*Kirschvink*, 1980] to define a consistent characteristic magnetization. The statistical methods of *Fisher* [1953] are reported for the single-component site means and overall paleomagnetic pole positions, and Bingham statistics [*Onstott*, 1980] were used to calculate the mean directions for those sites exhibiting intersecting demagnetization planes. Corrections for tilting, generally less than 10°, were made by simple rotations of the measured bedding planes to the horizontal about the strike axes.

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RESULTS

Sites Near La Purisima

Middle Miocene calc-alkaline basalt and andesite and middle and late Miocene alkalic basalt and basaltic andesite flows were sampled at 25 sites near La Purisima (Figure 1). K-Ar ages for a number of these flows were reported by *Sawlan and Smith* [1984] and are given along with estimated ages for the other flows in Table 1 of this study. Mostly flat lying, the alkalic flows form resistant caps on the sandstone and conglomerate mesas of the region, are generally 5–10 m thick and 5–10 km in length, and are usually associated in small groups with steep-sided, widely distributed cinder and spatter cones which fed the short flows. The calc-alkaline rocks are the eroded remnants of large volcanic edifices in which flows are subordinate to fragmental beds.

Intensities of natural remanent magnetization (NRM) for the samples range from approximately 4.9×10^{-1} to 9.0×10^{2} A m^{-1}. Those samples having higher NRM intensities also tend to show more dispersed NRM directions suggesting that the NRM at many sites has been affected by lightning strikes. Progressive af demagnetization was usually successful in removing lightning-induced magnetizations and defining a consistent higher-coercivity (generally >20 mT) component of magnetization (Figure 2*a*). Demagnetization planes were required at eight sites (Table 1) to determine the characteristic magnetizations, and an occasional sample was discarded which had been completely overprinted with a spurious magnetization. Thermal demagnetization experiments on samples from site LP25 define a single component of magnetization between 200° and 600°C (Figure 2*b*), similar to those isolated with af treatments. The consistent high-coercivity and high-blocking temperature component of remanent magnetization in the lava flows is inferred to have been acquired during their initial cooling.

Site mean directions for the lava flows near La Purisima are shown in Figure 3*a* and are listed in Table 1. Divided between both normal and reversed polarities, these directions group near the expected Miocene direction for North America with the exception of three sites (LP7, LP8, LP19). Site LP7 is in a flow immediately overlying the flow including site LP8, and site LP19 is situated in a nearby exposure most likely of one of the two flows. The VGP positions for these three sites are well beyond the limits of normal secular variation of the geomagnetic field (VGP latitude <45°), suggesting that these flows were erupted and magnetized over a brief interval of time during a polarity transition or geomagnetic excursion. Consequently, these three site mean VGP's have been excluded from the overall average for the Baja Peninsula (Table 2). The individual VGP's for the other flows near La Purisima are considered spot readings of the Miocene dipole field.

Sites Near San Ignacio

Early and middle Miocene andesites, late Miocene alkalic basalts, and the tholeiitic basalt of Esperanza were sampled at 17 sites in the vicinity of San Ignacio (Figure 1). The andesitic lava flows form coalescing networks of sinuous flows near the vent regions, are interbedded with abundant volcanic fragmental beds, and are shorter (2–3 km in length) and thicker (20–50 m thick) than the younger basaltic flows. The andesitic flows commonly dip 5° to 10°, and dips as steep as 18° were observed. These dips are considered primary because the lavas were emplaced on the flanks of composite volcanoes. The eroded andesitic deposits are locally overlain by generally flat-lying alkalic basalt and basaltic andesite flows similar to those found near La Purisima. Coeval with the alkalic basalts are the tholeiitic flows of Esperanza exposed in a narrow corridor extending from near the Gulf coast in the east to the Pacific coast in the west. In the easternmost exposures, numerous sinuous flows form a braided network, and sequences only a kilometer apart generally differ in chemistry and numbers of flows. Near San Ignacio the lavas are more sheetlike and laterally extensive. Individual flows may have traveled more than 200 km, and most flows are only between 3 and 8 m thick. A maximum of five flows was observed at any one locality. For some of the andesitic and basaltic lava flows near San Ignacio *Sawlan and Smith* [1984] reported K-Ar ages which are shown with estimated ages for the other flows in Table 1.

Stepwise af demagnetization of these rocks yields consistent magnetizations (Figure 2*c*). Site mean directions depicted in Figure 3*b* (Table 1) are near the expected Miocene North American normal and reversed-polarity directions with several notable exceptions. The directions for sites SI4, SI5, and SI8 are well beyond the limits of normal secular variation and were probably magnetized simultaneously under a transitional geomagnetic field. Sites SI4 and SI5 are from two different flows at the same locality, and site SI8 is from a flow at the same elevation 2 km to the east. The site mean VGP's for these three sites have been excluded from the mean paleomagnetic pole calculation for the Baja Peninsula. Site SI17 represents a resampling of one of *Pischke*'s [1979] sites, and the results from these sites will be discussed in a later section. Sites SI14, SI15, and SI16 were collected at a single locality in the basalt of Esperanza, have statistically indistinguishable directions, and probably represent a very short segment of time within the period of secular variation. These three site mean directions were averaged to form a single VGP used in the calculation of the mean pole position (Table 2).

Sites Near Bahia de La Paz

Samples were collected near Bahia de La Paz in early to middle Miocene rocks at 11 sites in the vicinity of San Juan de la Costa (BP1-11) and at four sites in the vicinity of La Paz (BP12-16, Figure 1; *Hausback* [1984*a*]). *Hausback* [1984*b*] determined K-Ar ages for many of these units (Table 1). Of the 15 sites total, 11 are in rhyolitic ash-flow tuffs or lava flows and 4 (BP3, BP9-11) are in thin sections of volcaniclastic sandstone and water-lain tuff. The units sampled represent primarily flat-lying deposits of different volcanic eruptions or sedimentary depositional periods. The La Paz tuff, a regional stratigraphic marker, was sampled in both the San Juan de al Costa and La Paz areas at sites BP4 and BP14, respectively.

Progressive af demagnetization of samples removed scattered components of magnetization at low coercivities (generally <20 mT) isolating a stable higher-coercivity (20–80 mT) component (Figure 2*d*). Thermal demagnetization treatments of representative specimens from each site yield similar results, with complete demagnetization being accomplished by 600°C. Petrographic observations of polished grain mounts made from heavy mineral separates of the ash-flow tuffs indicate that titanomagnetite is the predominant magnetic remanence carrier in these rocks. Site mean

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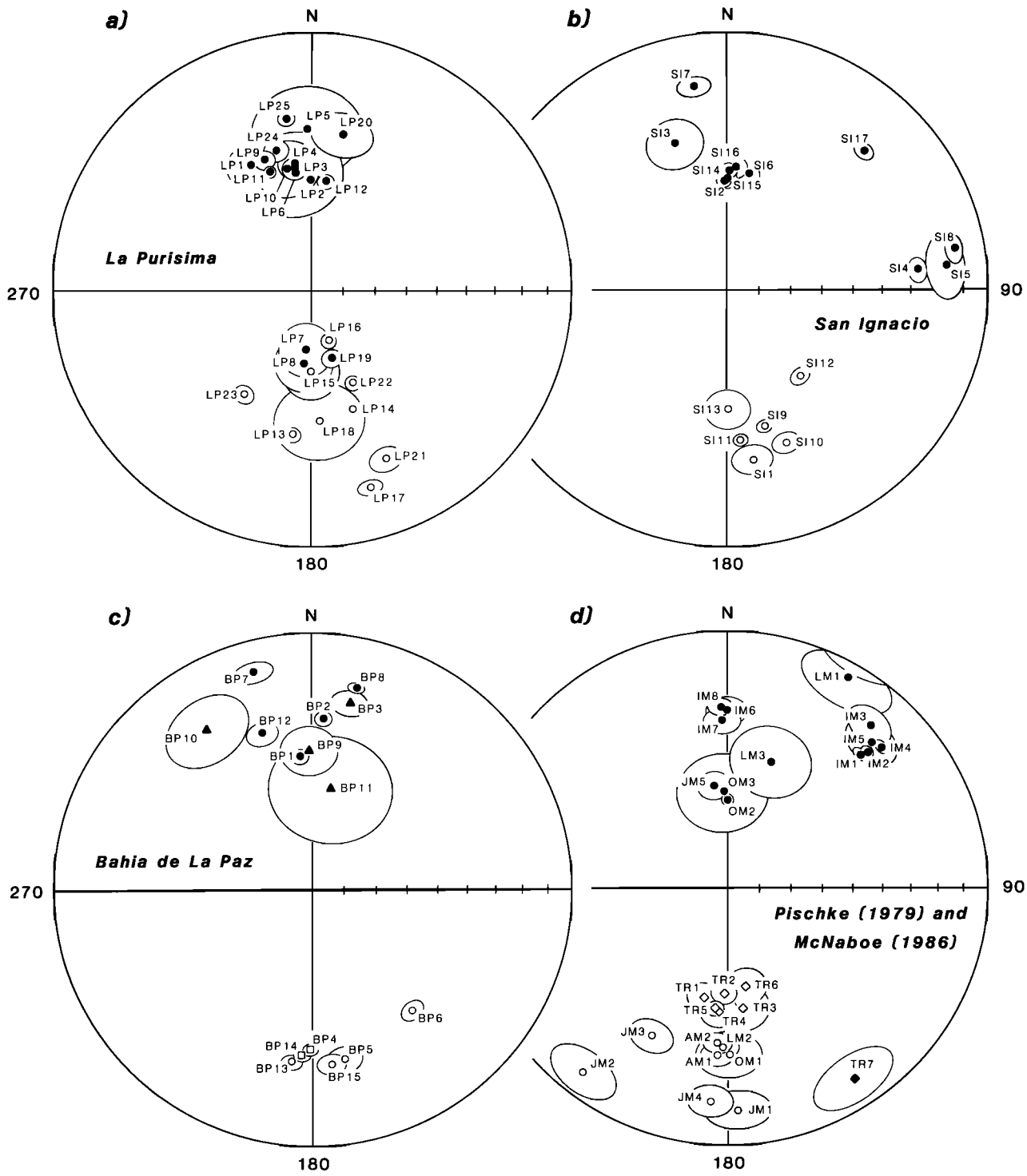


Fig. 3. Site mean directions with 95% confidence limits for (*a*) sites near La Purisima, (*b*) sites near San Ignacio, (*c*) sites near Bahia de La Paz (triangles indicate site means for sedimentary rocks, and squares indicate site means for the La Paz tuff), and (*d*) results of *Pischke* [1979] (circles) and *McNaboe* [1986] (diamonds). Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere and unfilled symbols from the upper hemisphere.

directions for the rocks near Bahia de La Paz (Figure 3*c*, Table 1) show both normal and reversed polarities and are generally similar to the expected North American direction. VGP's for the two sites in the La Paz tuff (BP4,14) are statistically indistinguishable and are averaged together before inclusion in the overall mean pole calculation (Table 2).

Results of Pischke [1979] and McNaboe [1986]

Pischke [1979] collected Miocene volcanic rocks for paleomagnetic study and K-Ar dating from 21 sites at five widely distributed localities on the Baja California Peninsula, and *McNaboe* [1986] sampled basalt flows in the Miocene Rosarito Beach Formation near Tijuana at seven

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TABLE 2. Paleomagnetic Poles for the Baja California Peninsula, Mainland Mexico, and North America

Region	N	R	K	A_{95} (dp/dm)	Plat	Plong	d	r	Reference
Miocene (Figure 4)									
Baja Peninsula	64	62.0968	22	3.8	86.4	142.1	−0.2 ± 3.8	−1.7 ± 4.3	this study
North American craton	18	17.8696	130	3.0	87.4	129.7			Irving and Irving [1982], including *Nairn et al.* [1975] and *Bobier and Robin* [1983] combined
Late Oligocene to Early Miocene (Figure 5a)									
Sierra Madre Occidental	17	16.4238	28	6.9	73.9	153.7	1.8 ± 6.3	−9.8 ± 6.7	*Nairn et al.* [1975]
Sierra Madre Occidental	19	17.3262	11	10.7	77.4	149.0	1.8 ± 8.9	−5.9 ± 9.5	*Bobier and Robin* [1983]
North American craton	102	94.1021	13	4.1	83.2	148.0			*Diehl et al.* [1983]
Late Cretaceous to Early Tertiary (Figure 5b)									
Sierra Madre Occidental	10	9.4211	16	12.7	87.0	61.6	9.8 ± 10.3	24.9 ± 11.0	*Bobier and Robin* [1983]
Baja Peninsula	8	7.9734	263	3.4	83.3	344.4	10.9 ± 3.8	32.0 ± 4.5	*Hagstrum et al.* [1985]
North American craton	5	4.9921	505	3.4	67.3	186.3			*Irving and Irving* [1982, Table 1, entries 39 and 42–45]
Late Triassic (Figure 5c)									
Western Sonora	3			7.9/11.7	74.8	106.2	−9.8 ± 7.3	0.3 ± 10.4	*Cohen et al.* [1981]
Northern Mexico	14			4.9/9.0	76.4	119.2	−12.8 ± 5.5	−2.8 ± 8.3	*Nairn* [1976]
North American craton	13	12.8000	70	5.0	66.0	93.0			*Irving and Irving* [1982]

N, number of poles analyzed; R, vector sum; K, precision parameter [*Fisher*, 1953]; A_{95}, radius of 95% confidence (in degrees); dp/dm, oval axes of 95% confidence (in degrees); Plat, Plong, latitude and east longitude of paleomagnetic pole, respectively (in degrees); d, r, poleward displacement and clockwise rotation of an average locality with respect to the North American craton, respectively (in degrees), with 95% confidence limits after *Beck* [1980] as modified by *Demarest* [1983].

sites (Figure 1). Samples were taken as field-drilled cores, were oriented with a magnetic compass, and were subjected to af demagnetization. Site means and VGP positions are recalculated for all sites in this work to correct for numerical errors. The site mean VGP's are averaged prior to calculating the overall paleomagnetic pole position of this study (Table 2) based on the field relationships, estimated ages, and paleomagnetic results reported by *Pischke* [1979] and *McNaboe* which are outlined below.

Two basalt flows dipping 3° to the west were sampled by *Pischke* [1979] at two sites (AM1-2) 6.5 km north of the village of Asuncion on the Vizcaino Peninsula (Figure 1). Whole-rock K-Ar ages of 8.1 ± 3.2 [*Pischke*, 1979] and 5.70 ± 0.20 Ma [*Sawlan and Smith*, 1984] were determined for the lowermost flow sampled for paleomagnetism at site AM1, and the flow underlying that sampled 2.5 km farther north at site AM2 is dated at 13.1 ± 3.2 Ma [*Throughton*, 1974]. These two flows were probably emplaced within a short period of time, as is suggested by their close proximity in the field, a lack of interlayered sediments, and similar paleomagnetic directions (Figure 3*d*, Table 1) and, consequently, their site mean VGP's have been averaged.

Pischke [1979] collected paleomagnetic samples at eight sites in flat-lying basalt and basaltic-andesite(?) flows near San Ignacio (Figure 1): five sites are in the basalt of Ezperanza (IM1-5), and three sites are in overlying basaltic-andesite flows most likely belonging to the alkalic unit of *Sawlan and Smith* [1984]. The basalt flow sampled at site IM1 is exposed by a roadcut approximately 3 km west of San Ignacio and was resampled in this study (SI17). An age of 6.1 ± 1.1 Ma for this flow determined by *Pischke* [1979] conflicts with ages for the basalt of Esparanza reported by *Gastil et al.* [1979] (11.1 ± 0.8 Ma) and *Sawlan and Smith* [1984] (10.1 ± 0.4 and 9.7 ± 0.3 Ma). Three flows collected at sites IM2-4 are in direct contact with each other and are exposed in a stream wash 500 m to the west of site IM1. The flow sampled at site IM2 is dated at 10.8 ± 6.4 Ma [*Pischke*, 1979]. A flow at roughly the same elevation was sampled across the valley 1 km southeast of San Ignacio at site IM5. The flows containing sites IM1, IM4, and IM5 directly overlie the Miocene San Ignacio Formation, and similar site mean directions for the five sites (IM1-5 including SI17) indicate that they are probably contemporaneous (Figure 3*d*, Table 1). Thus the VGP positions for sites IM1 through IM5 and SI17 have been averaged. The three sites in the younger alkalic basaltic-andesite flows are located in a section exposed 12 km west of San Ignacio and are dated (IM7) at 5.1 ± 0.2 Ma [*Pischke*, 1979]. The two flows including sites IM7 and IM8 are in direct contact and are separated by 40 m of breccia and brecciated flows from an upper flow (IM6). Because of the close field proximity and indistinguishable site mean directions (Figure 3*d*), these three flows are considered contemporaneous and their VGP's have been averaged.

Three basalt flows (JM1,3,4), an andesite flow (JM2), and an andesite dike (JM5) were sampled by *Pischke* [1979] at five sites west of Loreto in Baja California Sur (Figure 1). Sites JM1 and JM2 are significantly higher in the section than sites JM3 and JM4, and the entire section dips approximately 5° to the west. The basalt flows at sites JM1 and JM3 and the andesite flow at site JM2 are dated at 11.0 ± 7.7, 18.7 ± 3.4, and 14.3 ± 2.6 Ma [*Pischke*, 1979], respectively. The andesite dike at JM5 yields a K-Ar age of 7.6 ± 5.3 Ma [*Pischke*, 1979]; K-Ar determinations on most dikes in this region range from 15 to 19 Ma [*Gastil et al.*, 1979].

Near La Paz (Figure 1), *Pischke* [1979] sampled an andesite flow included in a sequence of volcaniclastic rocks dipping 43° to the northeast (LM1), a flat-lying ash-flow tuff (LM2) dated at 9.0 ± 0.7 Ma [*Pischke*, 1979], and an ash-flow tuff dipping 3° to the west (LM3) dated at 14.9 ± 0.3 Ma

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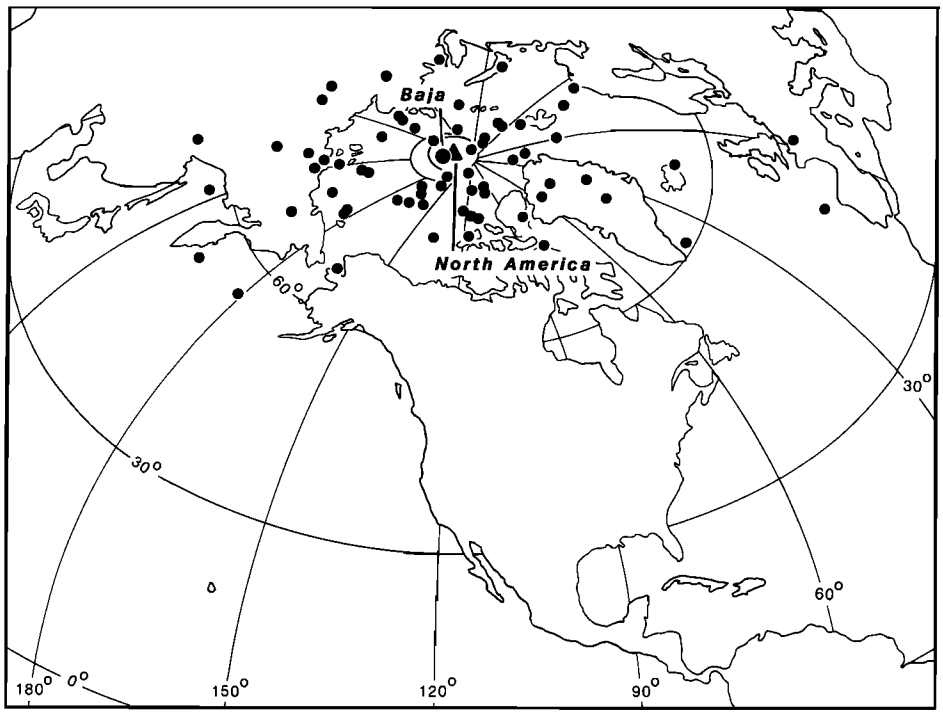


Fig. 4. Site mean and averaged Miocene virtual geomagnetic poles (Table 1) for the Baja California Peninsula. Larger symbols with 95% confidence limits indicate average paleomagnetic poles (Table 2) for the Baja Peninsula (circle) and the North American craton (triangle).

[*Pischke*, 1979]. A low Koenigsberger ratio [0.5; *Pischke*, 1979] and an in situ site mean direction near that of the present-day field indicates that the andesite flow sampled at site LM1 does not carry a stable remanent magnetization. Site LM1 has been excluded from the mean pole calculation of this study.

To the north in the state of Baja California, three flat-lying flows were sampled by *Pischke* [1979] in Lluway wash (Figure 1), where two adjacent ash-flow tuffs (sites OM2-3) overlie a weathered basalt flow (OM1) which, in turn, overlies a volcanic sandstone. The basalt yields an K-Ar age of 14.4 ± 1.9 Ma, and the uppermost ash-flow tuff yields an K-Ar age of 12.5 ± 0.9 Ma [*Pischke*, 1979]. The two ash-flow tuffs have similar site mean VGP's (Figure 3*d*) that have been averaged.

McNaboe [1986] sampled seven basalt flows in the gently dipping Miocene Rosarito Beach Formation [*Minch*, 1967] near Tijuana (Figure 1). A K-Ar age of 14.3 ± 2.6 Ma has been determined for a flow in the Costa Azul Member by *Hawkins* [1970]. Site TR1, TR2, and TR7 are in flows of this lower member, sites TR3 and TR4 are in the Las Glorias Member, and sites TR5 and TR6 are in the uppermost Los Buenos Member. Because of close geographic and stratigraphic relations between TR3 and TR4, their results (Figure 3*d*, Table 1) have been averaged. The VGP for site TR7 has an unacceptable VGP latitude (<45°) and has been excluded from the overall mean pole calculation.

Discussion and Conclusions

Dispersion of VGP's

The site mean and averaged VGP's for the Miocene rocks of the Baja California Peninsula (Table 1) are shown in Figure 4. The observed angular standard deviation of these points after structural correction is 17.3°. This overall dispersion is due primarily to contributions from errors in the experimental procedure, incomplete structural corrections, geomagnetic secular variation, and within-site dispersion. Using the site level precision parameters as estimates of within-site dispersion, this contribution can be removed from the overall value leaving an angular standard deviation of 16.9° for the combined effects of the remaining sources of dispersion. This value is significantly greater (at the 95% confidence level) than a model value for dispersion due to secular variation (15.0°) at an average Miocene paleomagnetic latitude for the Baja Peninsula [*McFadden*, 1980]. A similar conclusion is drawn from a comparison of the between-site dispersion of this study with the amplitude of secular variation for the past 5 m.y. estimated by *McFadden and McElhinny* [1984] to be 15.1° with lower and upper 95% confidence limits of 14.4° and 15.9° for latitudes of the Baja Peninsula. Thus significant errors likely arise from either the structural corrections or experimental procedure, or both.

The sources of dispersion in the peninsular California dataset may be evaluated further by analyzing the shape of the distribution of VGP's. The VGP mapping function introduces an elongation to a symmetrical set of points so that a symmetrical set of directions maps to a distribution of VGP's elongate in the direction normal to the plane of the paleomagnetic meridian and, conversely, a symmetrical set of VGP's maps to a distribution of directions elongate in the vertical plane [e.g., *Cox*, 1970]. A calculation of the two principal moments of inertia for the Miocene VGP's for the Baja Peninsula [*Engebretson and Beck*, 1978] shows that they are not circularly symmetrical (axial ratio = 1), but are elongate (axial ratio = 1.74) along a plane with a pole at 1.2°N, 250.4°E. This elongation is perpendicular to the paleomagnetic meridian for the Baja Peninsula and is larger

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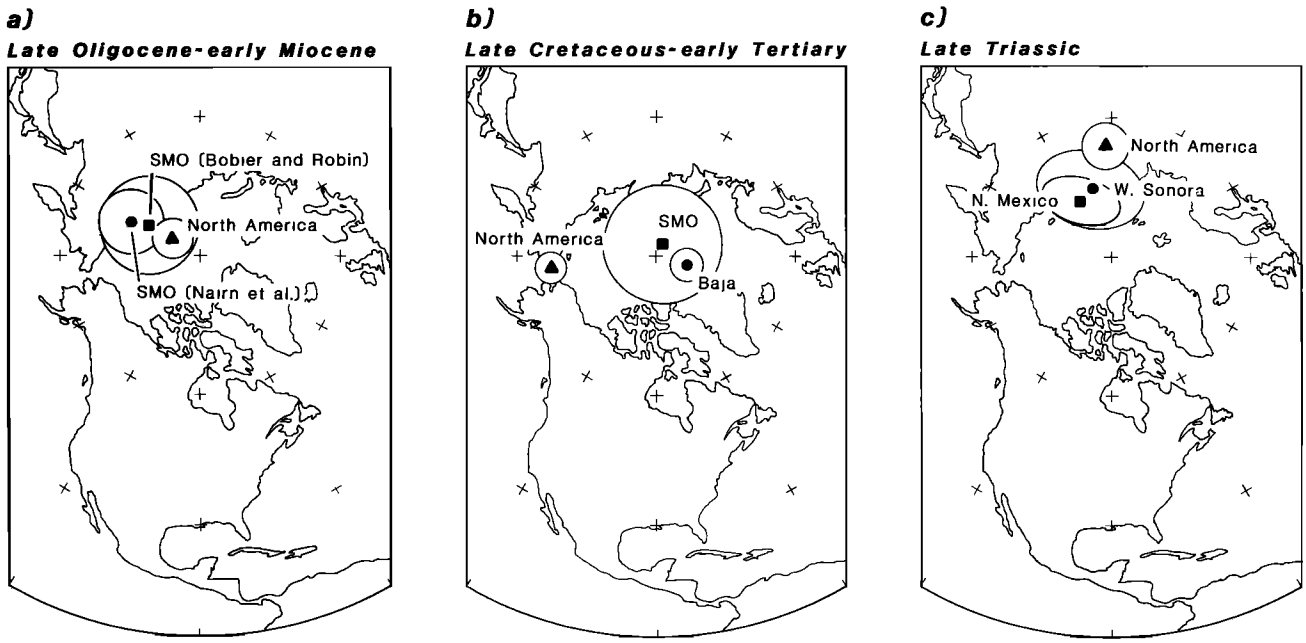


Fig. 5. Paleomagnetic poles with 95% confidence limits (Table 2) for (*a*) late Oligocene to early Miocene rocks of the Sierra Madre Occidental (SMO) determined by *Nairn et al.* [1975] (circle) and *Bobier and Robin* [1983] (square) and of the United States of America and Canada reported by *Diehl et al.* [1983] (triangle); (*b*) Late Cretaceous to early Tertiary rocks of the Baja California Peninsula [*Hagstrum et al.*, 1985] (circle), of the Sierra Madre Occidental (SMO) near Mazatlan, Sinaloa [*Bobier and Robin*, 1983] (square), and of the North American craton [*Irving and Irving*, 1982] (triangle); and (*c*) Late Triassic rocks of western Sonora [*Cohen et al.*, 1981] (circle), of northern Mexico [*Nairn*, 1976] (square), and of the North American craton [*Irving and Irving*, 1982] (triangle).

than that which can be attributed solely to the VGP mapping function (1.46–1.67 for inclinations typical of the Baja Peninsula). This suggests that the sources of dispersion other than secular variation are associated with the declination of directions and would most likely be caused by undetected structural rotations about vertical axes or misorientations of the field-drilled cores in the horizontal plane. Because structural corrections in this study are generally small, it seems probable that greater error was introduced in the field orientations of the sample cores. A suncompass was used to orient the strike of the cores at sampling sites near La Purisima, and back azimuths were routinely made at sites near San Ignacio. However, only single magnetic azimuths were taken for samples at the other sites, and magnetic orientations can prove unreliable in strongly magnetic volcanic rocks.

Tectonic Implications

The average paleomagnetic pole positions for Miocene rocks from the Baja Peninsula and the North American craton (Table 2) are also shown with their 95% confidence limits in Figure 4. The pole for the North American craton is the 20-Ma mean, 30-Ma window paleopole from Table 2 of the *Irving and Irving* [1982] compilation modified by the inclusion of an additional pole calculated from two studies in the Sierra Madre Occidental of western Mexico (Table 2). The 30-Ma window of the 20-Ma paleopole of *Irving and Irving* [1982] spans an interval of time equivalent to the range of ages for the rocks of this study (Table 1) and includes six poles from studies in mainland Mexico. This 20-Ma paleopole, however, does not include the work of *Nairn et al.* [1975] and *Bobier and Robin* [1983] from the southern Sierra Madre Occidental. Average VGP's for these two studies in general are not significantly different (Table 2) from a late Oligocene to early Miocene pole for rocks from the United States and Canada [*Diehl et al.*, 1983; Figure 5*a*], indicating that little or no relative motion has occurred between these localities since early Miocene time. Because *Nairn et al.* [1975] and Bobier and Robin sampled the younger rhyolitic volcanic rocks of the Sierra Madre Occidental along the road from Durango to Mazatlan, perhaps with some overlap, the results from these two studies have been combined to form a single VGP that has been added to those of the 20-Ma paleopole of Irving and Irving to make the Miocene paleopole for cratonic North America used in this study.

From the close proximity of the North American and peninsular California paleopoles shown in Figure 4 it is clear that the Baja Peninsula was contiguous with North America during Miocene time and not 10° south of its present position, as was proposed by *Pischke et al.* [1979] and *Marshall et al.* [1979]. This estimate of more southerly Miocene paleolatitudes for the Baja Peninsula was arrived at by Pischke through incorrect averaging of a smaller dataset (included in Table 1) by unspecified means. A comparison of the paleopoles for the Baja Peninsula and the North American craton (Table 2) suggests that the Baja Peninsula was approximately 0.2° farther north than its present position with respect to North America in Miocene time. This result indicates only that the translation of the Baja Peninsula associated with the Pliocene opening of the Gulf of California is within the error limits of the paleomagnetic data. In Figure 6 the error limits in latitude for the northern and southern tips of peninsular California are shown to include the extent of this most recent tectonic movement. The Miocene paleomagnetic data for the Baja Peninsula also do

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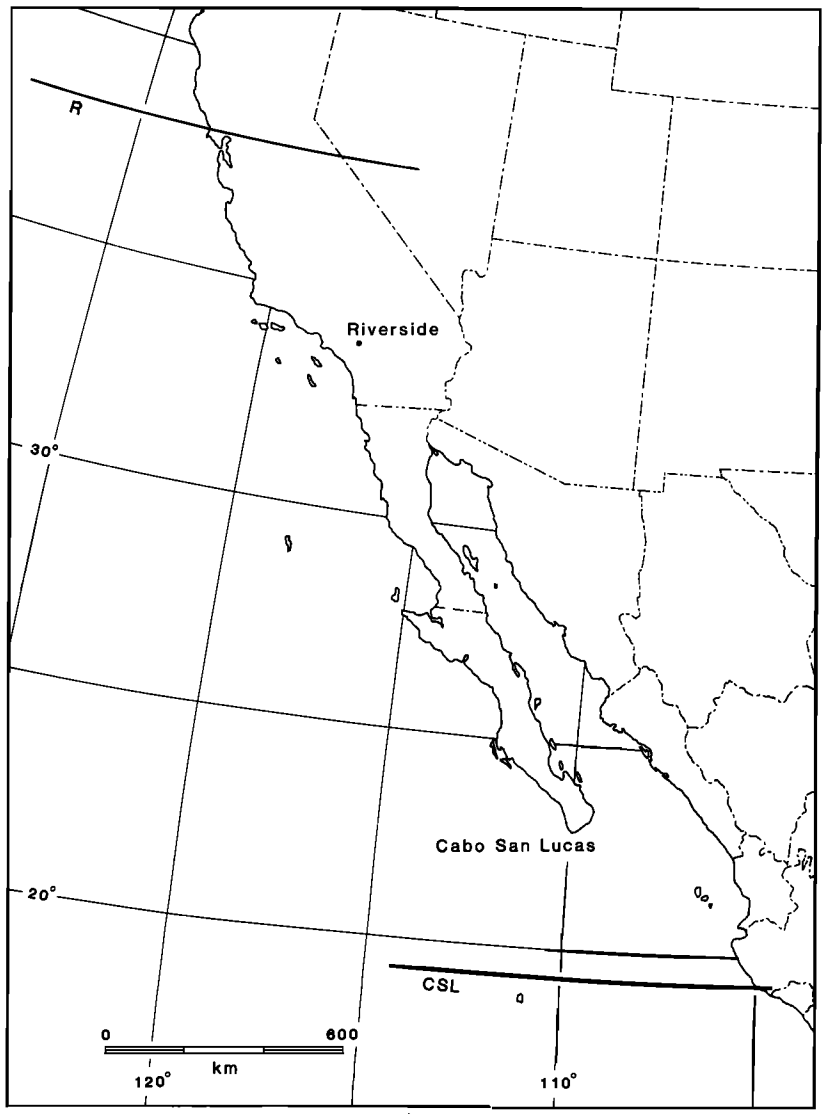


Fig. 6. Map of southwestern North America showing the northernmost (R) and southernmost (CSL) 95% confidence limits for the northern tip (Riverside, California) and southern tip (Cabo San Lucas, Baja California Sur) of peninsular California for Miocene time.

not support a shallow-inclination geomagnetic field in the region during this time. As is mentioned by *Hagstrum et al.* [1986], the evidence for persistent shallow inclinations in southern California and western Arizona [*Luyendyk et al.*, 1985] is also unconvincing. Recently acquired results for early Miocene rocks from the Ajo mining district, Arizona, are consistent with those from the North American craton [*Hagstrum et al.*, 1987] and do not support the late Miocene shallow inclinations or counterclockwise rotation for southwestern Arizona reported by *Calderone and Butler* [1984].

The paleomagnetic results of this study affirm that the major portion of northward motion (9° of latitude, 24° of clockwise rotation) of the Baja Peninsula since Late Cretaceous to early Tertiary time [e.g., *Hagstrum et al.*, 1985] occurred prior to early Miocene time. Similarities between the Late Cretaceous to early Tertiary basement rocks of the Baja Peninsula and mainland Mexico across the Gulf of California [*Gastil et al.*, 1976; *Gastil and Krummenacher*, 1977; *Silver*, 1986] suggest that these rocks of mainland Mexico are an eastern, although somewhat younger, continuation of the Peninsular Ranges batholith. This correlation implies that if peninsular California had been displaced 9° northward in early Tertiary time, the locus of motion between it and North America was to the east of the present Gulf of California. Paleomagnetic results [*Bobier and Robin*, 1983] from the basement (102 to 45 Ma) rocks of the Sierra Madre Occidental near Mazatlan, Sinaloa (Figure 1) are similar to those [*Hagstrum et al.*, 1985] from the Baja Peninsula (Figure 5*b*, Table 2) supporting this correlation across the Gulf as well as the early Tertiary northward translation of peninsular California. Preliminary paleomagnetic data [*Cohen et al.*, 1981] from the Upper Triassic to Lower Jurassic Antimonio Formation exposed 50 km west of Caborca, Sonora (Figure 1) are similar to other coeval results from north-central Mexico [*Nairn*, 1976], and indicate, if any movement at all, a southward displacement with respect to the United States and Canada (*Irving and Irving* [1982]; Figure 5*c*, Table 2). These Triassic-Jurassic results

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imply that the early Tertiary boundary between peninsular California and North America runs approximately northwest-southeast between Mazatlan and the site west of Caborca. Reconnaissance mapping of western Sonora by *Merriam* [1972] and *Gastil and Krummenacher* [1977] delineates a number of northwest trending right-lateral strike-slip faults, including the craton margin lineament shown in Figure 1. One or more of these faults may represent the early Tertiary boundary between peninsular California and the North American craton, although, on the basis of geochemical and geochronologic data, *Silver* [1986] contends that the locus of intrusive activity related to the Peninsular Ranges batholith migrated eastward across the craton margin of Sonora, without offset, reaching eastern Sonora by about 60 Ma.

Northward movement of the Late Cretaceous to early Tertiary rocks of western Sinaloa in Paleogene time, indicated by the paleomagnetic data reported by *Bobier and Robin* [1983], suggests that the age gap between the intermediate-composition volcanic and intrusive basement rocks (>45 Ma) and younger rhyolitic volcanic rocks (<34 Ma) of the Sierra Madre Occidental represents a tectonic unconformity rather than a hiatus in igneous activity as suggested by *McDowell and Clabaugh* [1979]. Moreover, the time represented by this age gap is likely the time of tectonic transport. Paleomagnetic data from sedimentary rocks on San Miguel Island off southern California [*Champion et al.*, 1986], as young as early to middle Eocene in age (approximately 55 to 42 Ma), also support the Paleogene (45 to 34 Ma) northward movement of peninsular California. The 45-Ma age for the approximate start of tectonic movement generally corresponds in time to the plate reorganization associated with the bend in the Hawaiian-Emperor seamount chain [*Dalrymple and Clague*, 1976]. Although the fixed hot spot model of *Engebretson et al.* [1986] shows no significant change in convergence direction between the Farallon and North American plates at 45 Ma, relative plate motion models indicate a significant increase in the northward tangential component of convergence (especially if Antarctica is treated as two plates) at approximately this time [*Jurdy*, 1984].

The available paleomagnetic data from western Mexico and southern California indicate that during early Tertiary time the Baja California Peninsula and a portion of Sinaloa and Sonora were displaced northward and rotated clockwise with respect to cratonic North America along a plate boundary which lies to the east of the present Gulf of California. This movement was perhaps initiated with the shift in motion of the Pacific and Farallon plates relative to North America at approximately 45 Ma and ended by late Oligocene to early Miocene time. The Baja Peninsula remained in position against the Mexican mainland until about 6 Ma, when it was moved northwestward with the rifting and opening of the Gulf of California.

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